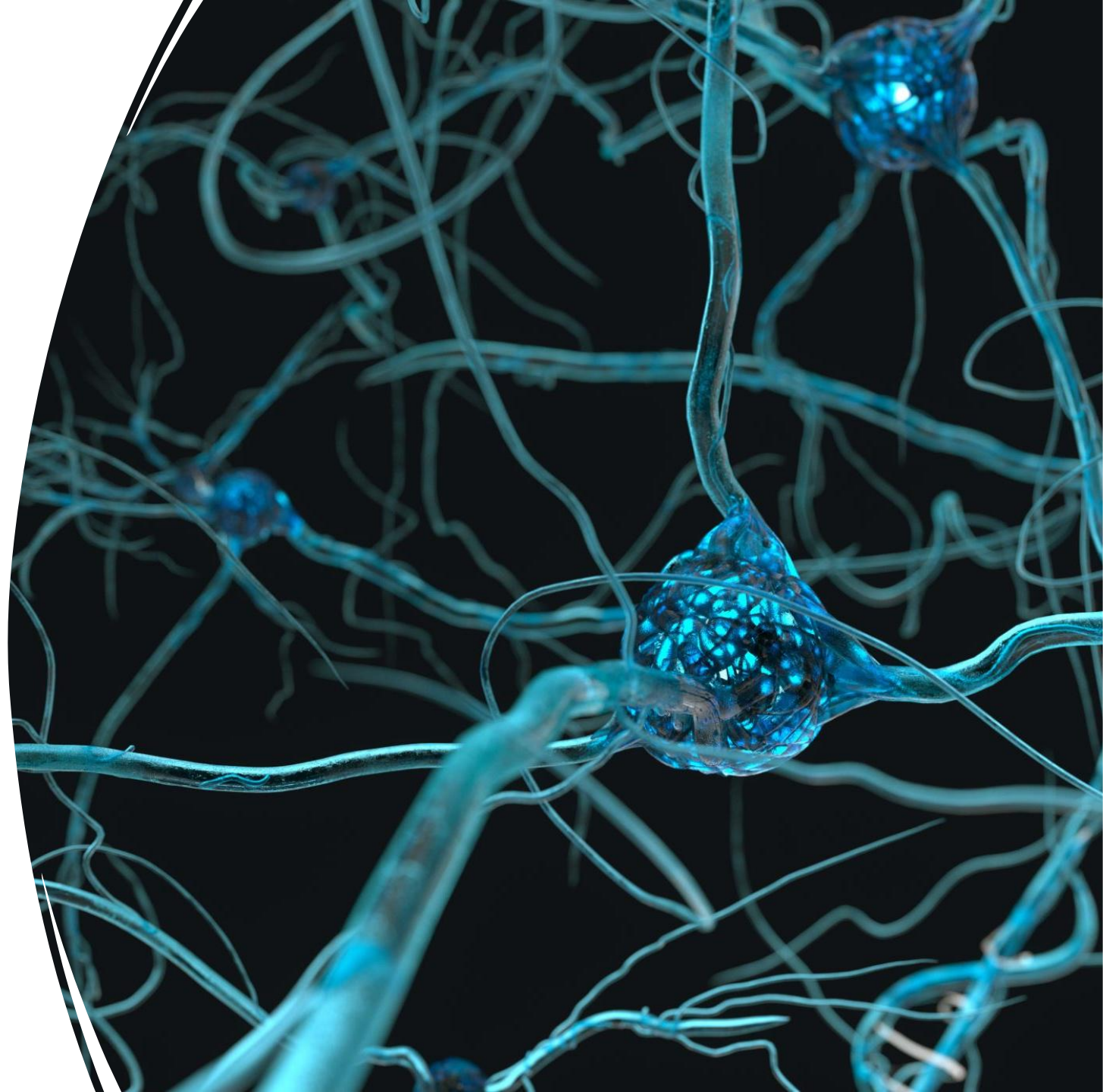


Fong Hei Law

MSc Brain Sciences Project

Supervisor: Dr Junichi Haschisuka

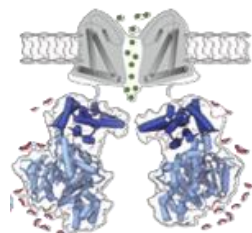
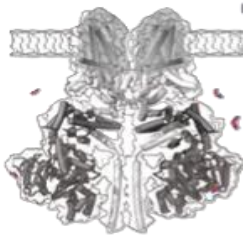
Characterization of spinal cord lamina I projection neurons: electrophysiological responses to cutaneous stimuli



How does the nervous system sense cold?

Cold stimuli activates primary neurons and projection neurons

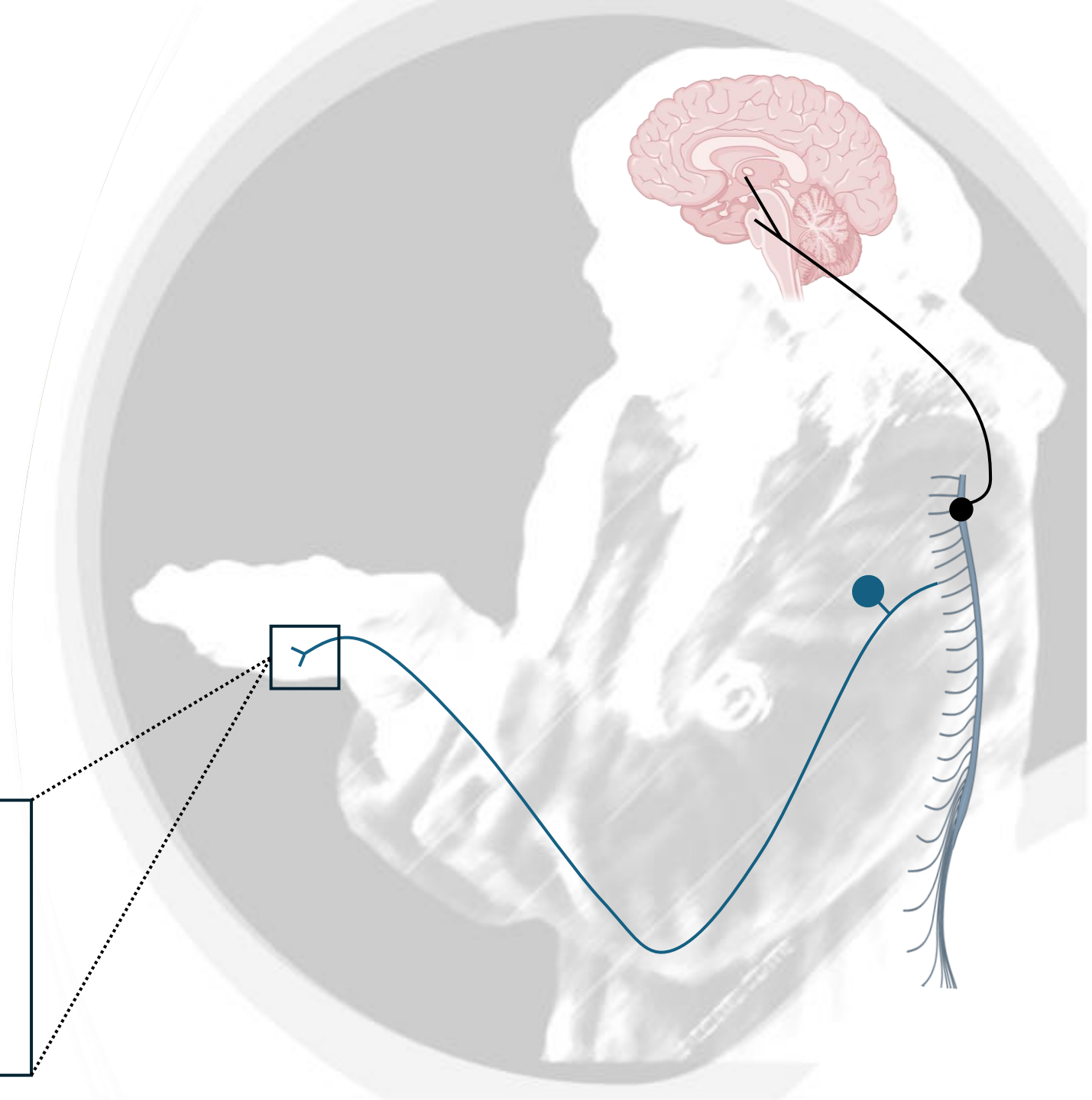
Skin cooling activates Transient Receptor Potential Member 8 (TRPM8) of primary afferents that synapse onto a subtype of lamina I projection neurons (ALS3) responsible for relaying the cold signal to the brain (exclusively)



TRPM8



Action
Potential



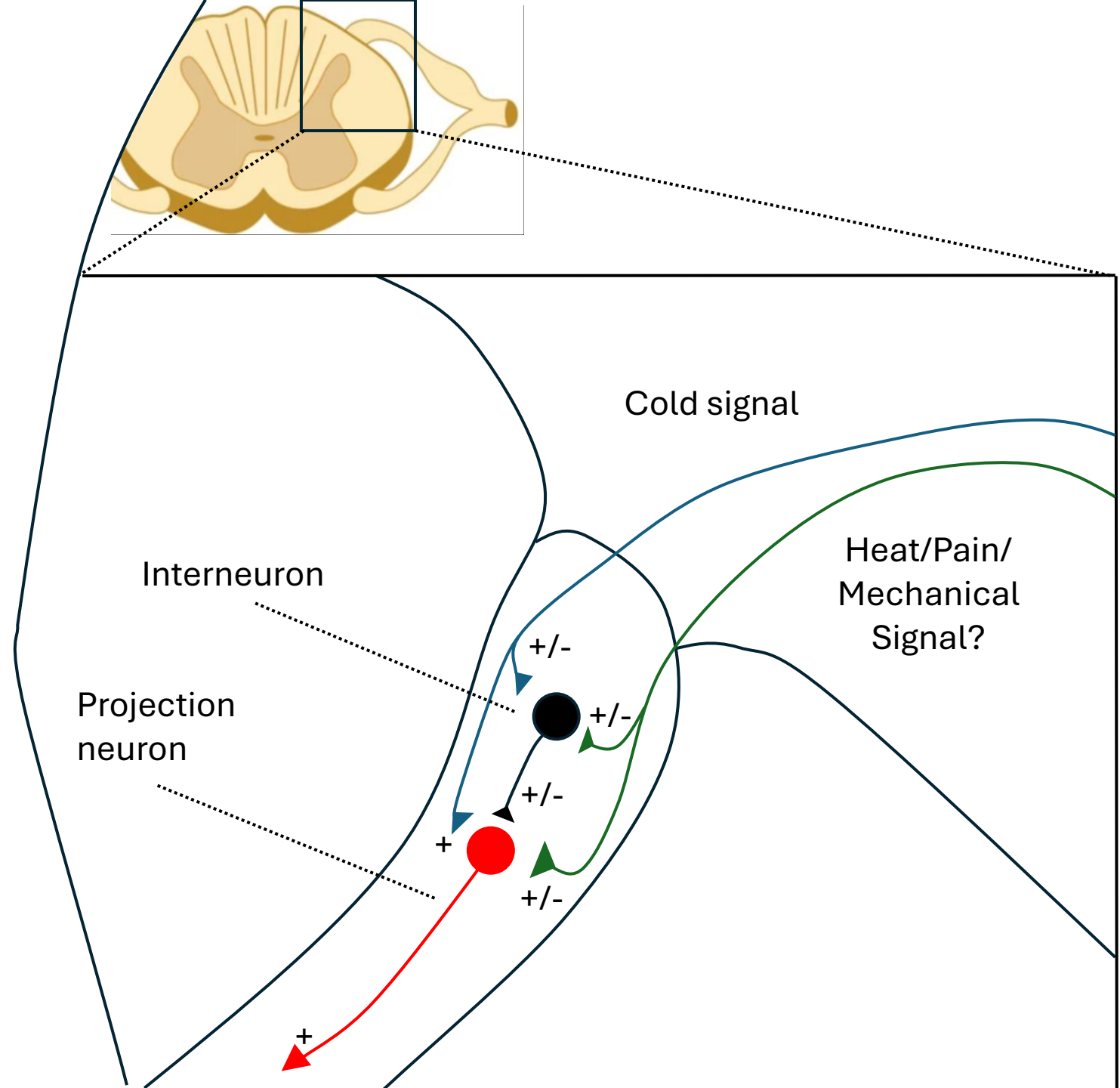
Characterization of projection neurons by action potential is not enough

Interneurons modulate signals:

TRPM8 afferents exclusively activate some excitatory interneurons, and some inhibitory neurons (e.g. Nos1) provides input to other projection neurons but not cold-sensing neurons

Modalities often converge onto the same projection neuron

Projection neurons are sparse (1% of all spinal neurons), cold-sensing projection neurons are even more sparse because most projection neurons are polymodal



Aim: Characterize projection neurons by subthreshold inputs to understand spinal microcircuits



1. Prepare semi-intact ex-vivo preparation



2. Apply cold/hot saline/filament onto the skin



3. Record either EPSC or IPSC of lamina I projection neurons with voltage clamp



4. Repeat 4 times for each stimulus



5. Use Easy Electrophysiology to calculate frequency and amplitude of EPSC/IPSC

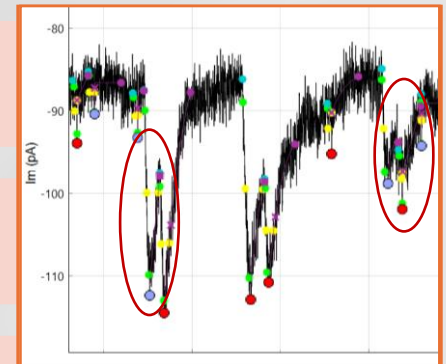
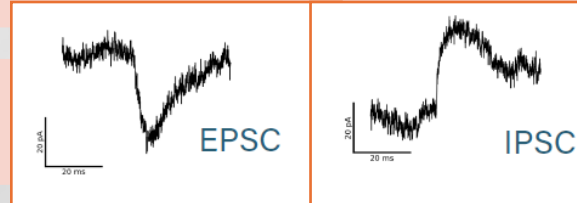
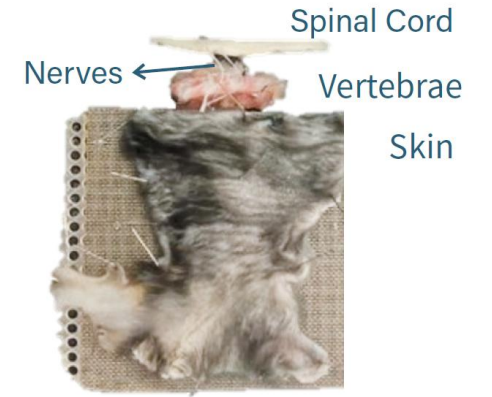


6. Count as a responder if frequency or amplitude after stimulation $\geq$ baseline + 3SD for at least two repetition

Phox2a::Cre
Crossed Mice



Semi-intact ex vivo preparation



Aim: Characterize projection neurons by subthreshold inputs to understand spinal microcircuits



1. Prepare semi-intact ex-vivo preparation



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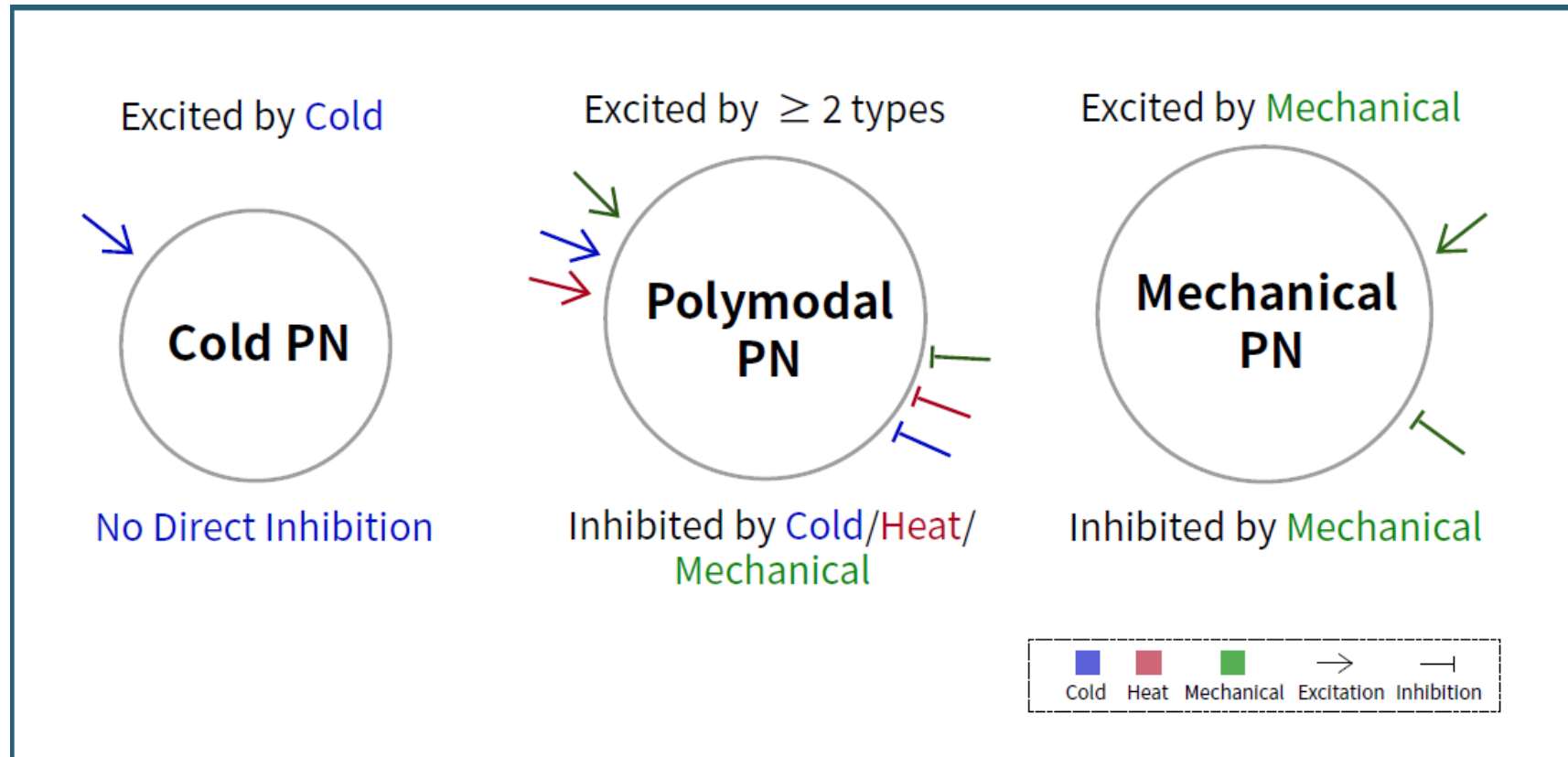
5. Use Easy Electrophysiology to calculate frequency and amplitude of EPSC/IPSC



6. Count as a responder if frequency or amplitude after stimulation $\geq$ baseline + 3SD for at least two repetition

Hypotheses:

Inhibitory inputs are consistent within subtypes of projection neurons (based on in-house preliminary data)

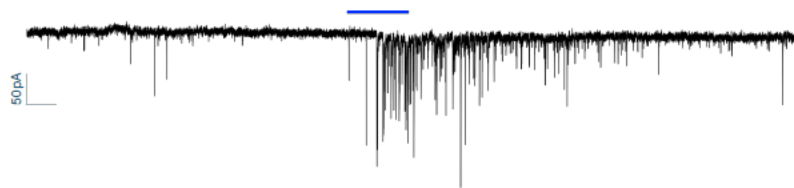


Example traces

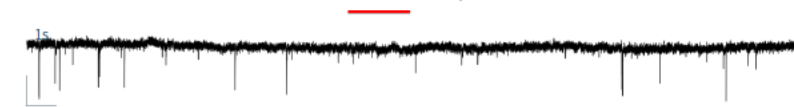
EPSC; $V_{\text{holding}} = -70\text{mV}$

IPSC; $V_{\text{holding}} = 0\text{mV}$

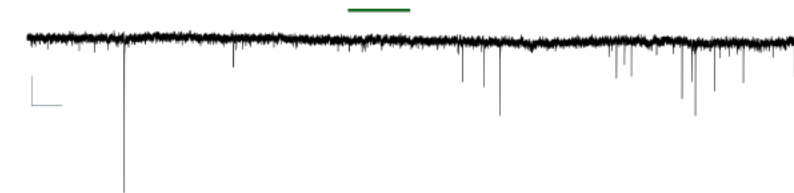
Cold Saline



Hot Saline



Von Frey 10g
(Mechanical)



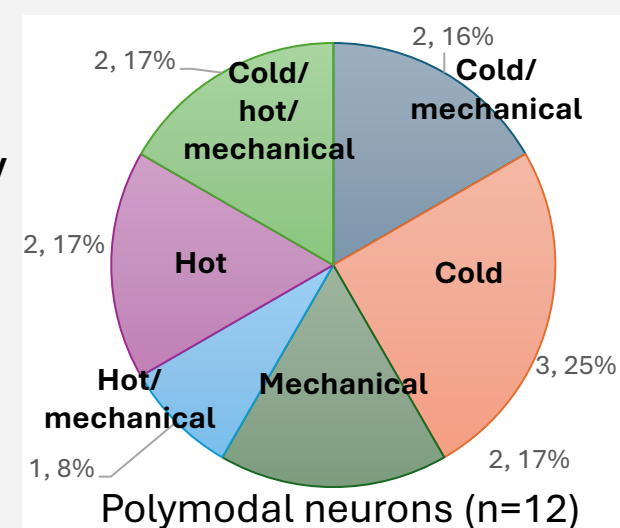
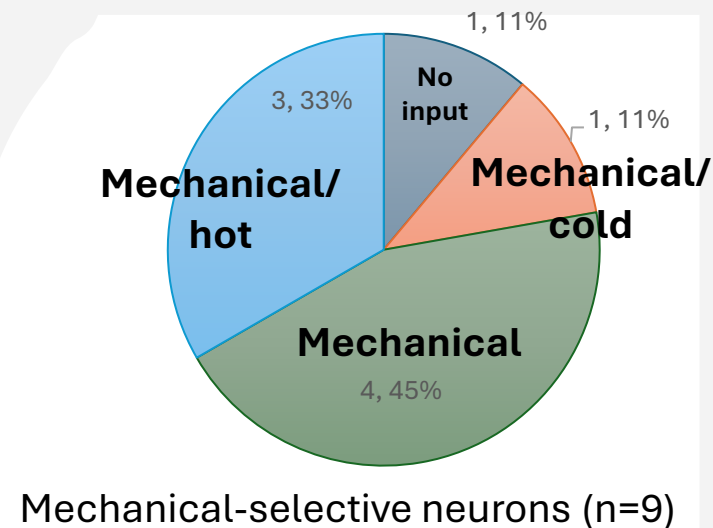
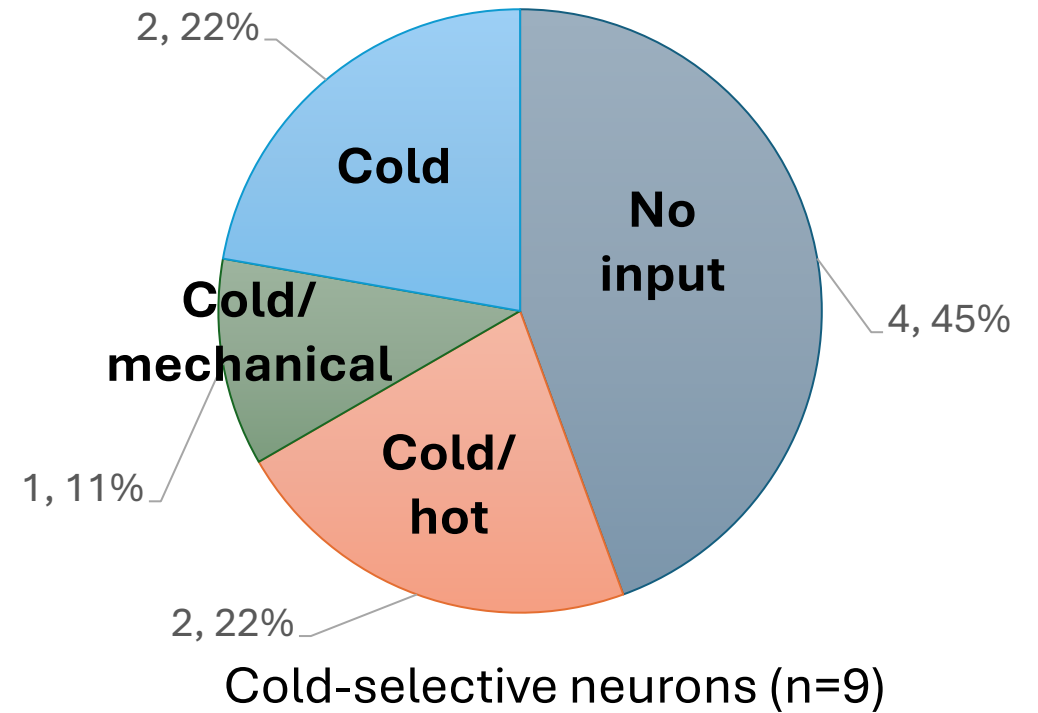
EPSCs to cold stimuli only
→ cold-selective PNs

Unable to evoke IPSCs
in cold-selective PNs

Proportion of inhibitory inputs by neuronal subtypes

- 1 Surprisingly, $\geq 50\%$ of cold neurons received direct inhibitory inputs, and those inputs include cold stimuli
- 2 Inhibitory inputs of mechanical neurons include mechanical stimuli, as expected
- 3 Inhibitory inputs of polymodal neurons are diverse, as expected

30 neurons from 16 animals,
112 recordings collected between 2023-2026
R Studio; Python; Easy Electrophysiology



Conclusion

Inhibitory inputs are consistent within each functionally defined spinal cord lamina I projection neurons

